

2.3 HTML Form: The <form> Element, Input Elements, Label and Field set

2.3.0 HTML Form

Forms add the ability to web pages to not only provide the person viewing the document with dynamic information but also to obtain information from the person viewing it, and process the information.

Objectives: Upon completing this section, you should be able to

- Create a FORM.
- Add elements to a FORM.
- Define CGI (Common Gateway Interface).
- Describe the purpose of a CGI Application.
- Specify an action for the FORM.
- Forms work in all browsers.
- Forms are Platform Independent.

To insert a form we use the <FORM></FORM> tags. The rest of the form elements must be inserted in between the form tags.

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Form</title>
</head>
<body>
  <!--form start-->
  <form name="" method="" action="" target="">
  </form>
  <!--form end-->
</body>
</html>
```

2.3.1 Form Element Attributes

- **ACTION:** It is the URL of the CGI (Common Gateway Interface) program that is going to accept the data from the form, process it, and send a response back to the browser.
- **METHOD:** GET (default) or POST specifies which HTTP method will be used to send the form's contents to the web server. The CGI application should be written to accept the data from either method.
- **NAME:** It is a form name used by VBScript or JavaScript.
- **TARGET:** It is the target frame where the response page will show up.

2.3.2 Form Input Elements

HTML form input elements are interactive controls used within an HTML <form> element to collect user input.

- Form elements have properties: Text boxes, Password boxes, Checkboxes, Option (Radio) buttons, Submit, Reset, File, Hidden and Image.
-

The properties are specified in the TYPE Attribute of the HTML element <INPUT></INPUT>.

```
<!DOCTYPE html>

<html lang="en">

  <head>

    <meta charset="UTF-8" />

    <meta name="viewport" content="width=device-width, initial-scale=1.0" />

    <title>HTML Form</title>

  </head>

  <body>

    <!--form start-->

    <form name="" method="" action="" target="">

      <input

        type="" name="" value="" checked

        size="" maxlength="" minlength="" required />

    </form> <!--form end-->

  </body></html>
```

<INPUT> Element's Properties	
TYPE	Type of INPUT entry field
NAME	Variable name passed to CGI application
VALUE	The data associated with the variable name to be passed to the CGI application
CHECKED	Button/box checked
SIZE	Number of visible characters in text field
MAXLENGTH	Maximum number of characters accepted
MINLENGTH	Minimum number of characters accepted

2.33 Form Input Elements, Label and Fieldset

Input Type	Description
<u><label></u>	It defines labels for <form> elements.
<u><input></u>	It is used to get input data from various types such as text, password, email, etc by changing its type.
<u><input type="text"></u>	Defines a one-line text input field
<u><input type="password"></u>	Defines a password field
<u><input type="submit"></u>	Defines a submit button
<u><input type="reset"></u>	Defines a reset button
<u><input type="radio"></u>	Defines a radio button
<u><input type="email"></u>	Validates that the input is a valid email address.

Input Type	Description
<u><input type="number"></u>	Allows the user to enter a number. You can specify min, max, and step attributes for range.
<u><input type="checkbox"></u>	Used for checkboxes where the user can select multiple options.
<u><input type="date"></u>	Allows the user to select a date from a calendar.
<u><input type="time"></u>	Allows the user to select a time.
<u><input type="file"></u>	Allows the user to select a file to upload.
<u><button></u>	It defines a clickable button to control other elements or execute a functionality.
<u><select></u>	It is used to create a drop-down list.
<u><textarea></u>	It is used to get input long text content.
<u><fieldset></u>	It is used to draw a box around other form elements and group the related data.
<u><legend></u>	It defines a caption for fieldset elements
<u><datalist></u>	It is used to specify pre-defined list options for input controls.
<u><output></u>	It displays the output of performed calculations.

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Form</title>
</head>
<body>
  <form>
    <fieldset>
      <legend>User Personal Information</legend>
      <label for="name">Enter your full name:</label>
      <input type="text" id="name" name="name" required /><br/>
      <label for="email">Enter your email:</label>
      <input type="email" id="email" name="email" required /><br/>
      <label for="password">Enter your password:</label>
      <input type="password" id="password" name="pass" required /><br/>
      <label for="confirmPassword">Confirm your password:</label>
      <input type="password" id="confirmPassword" name="confirmPass" required /> <br/>
    <select>
      <!-- option tag starts -->
      <option>Choose an option</option>
      <option value="html">HTML</option>
      <option value="java">JAVA</option>
      <option value="C++">C++</option>
      <option value="php">PHP</option>
      <!-- option tag ends -->
    </select>
  </form>
</body>
</html>

```

```
<label>Enter your gender:</label>

  <div class="gender-group">

    <input type="radio" name="gender" value="male" id="male" required />

    <label for="male">Male</label>

    <input type="radio" name="gender" value="female" id="female" />

    <label for="female">Female</label>

    <input type="radio" name="gender" value="others" id="others" />

    <label for="others">Others</label>

  </div>
<br/>

  <label for="dob">Enter your Date of Birth:</label>

  <input type="date" id="dob" name="dob" required /><br/>

  <label for="address">Enter your Address:</label>

  <textarea id="address" name="address" required></textarea>

<br/>

  <input type="submit" value="Submit" />

</fieldset>

</form>
</body>
</html>
```

2.4 HTML Form: Select and Option Tags, TextArea, Button Element

Input Type	Description
<code><select></code>	It is used to create a drop-down list.
<code><option></code>	It is used to define options in a drop-down list.
<code><optgroup></code>	It is used to define group-related options in a drop-down list.
<code><textarea></code>	It is used to get input long text content.
<code><input type="submit"></code>	Defines a submit button
<code><input type="reset"></code>	Defines a reset button

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>HTML Form</title>
</head>
<body>
  <form>
    <h2>HTML optgroup Tag</h2>
    <select>
```

```

<select>
  <!-- optgroup tag starts -->
  <optgroup label="Programming Languages">
    <option value="C">C</option>
    <option value="C++">C++</option>
    <option value="Java">Java</option>
  </optgroup>

  <optgroup label="Scripting Language">
    <option value="JavaScript">JavaScript</option>
    <option value="PHP">PHP</option>
    <option value="Shell">Shell</option>
  </optgroup>
  <!-- optgroup tag ends -->
</select>
</form>
</body>
</html>

```

See above example under 2.4 for textarea and button.

2.5 HTML Form: Form Attributes, Input Validation, Accessibility and Usability

2.5.0 Form Attribute See under 2.4 titled Form Attribute.

Attributes that are unique to particular input types—or attributes which are common to all input types but have special behaviors when used on a given input type—are instead documented on those types' pages.

Attributes for the <input> element include the global HTML attributes and additionally:

Attribute	Type(s)	Description
accept	file	Hint for expected file type in file upload controls
alt	image	alt attribute for the image type. Required for accessibility
autocapitalize	all except url, email, and password	Controls automatic capitalization in inputted text.
autocomplete	all except checkbox, radio, and buttons	Hint for form autofill feature
capture	file	Media capture input method in file upload controls
checked	checkbox, radio	Whether the command or control is checked
dirname	hidden, text, search, url, tel, email	Name of form field to use for sending the element's directionality in form submission
disabled	all	Whether the form control is disabled
form	all	Associates the control with a form element
formaction	image, submit	URL to use for form submission
formenctype	image, submit	Form data set encoding type to use for form submission
formmethod	image, submit	HTTP method to use for form submission
formnovalidate	image, submit	Bypass form control validation for form submission
formtarget	image, submit	Browsing context for form submission
height	image	Same as height attribute for ; vertical dimension
list	all except hidden, password, checkbox, radio, and buttons	Value of the id attribute of the <datalist> of autocomplete options

<u>max</u>	date, month, week, time, datetime-local, number, range	Maximum value
<u>maxlength</u>	text, search, url, tel, email, password	Maximum length (number of characters) of value
<u>min</u>	date, month, week, time, datetime-local, number, range	Minimum value
<u>minlength</u>	text, search, url, tel, email, password	Minimum length (number of characters) of value
<u>multiple</u>	email, file	Boolean. Whether to allow multiple values
<u>name</u>	all	Name of the form control. Submitted with the form as part of a name/value pair
<u>pattern</u>	text, search, url, tel, email, password	Pattern the value must match to be valid
<u>placeholder</u>	text, search, url, tel, email, password, number	Text that appears in the form control when it has no value set
<u>popovertarget</u>	button	Designates an <input type="button"> as a control for a popover element
<u>popovertargetaction</u>	button	Specifies the action that a popover control should perform

<u>readonly</u>	all except hidden, range, color, checkbox, radio, and buttons	Boolean. The value is not editable
<u>required</u>	all except hidden, range, color, and buttons	Boolean. A value is required or must be checked for the form to be submittable
<u>size</u>	text, search, url, tel, email, password	Size of the control
<u>src</u>	image	Same as src attribute for <u></u> ; address of image resource
<u>step</u>	date, month, week, time, datetime-local, number, range	Incremental values that are valid
<u>type</u>	all	Type of form control
<u>value</u>	all except image	The value of the control. When specified in the HTML, corresponds to the initial value
<u>width</u>	image	Same as width attribute for <u></u>

```
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>HTML Form</title>
  </head>
  <body>
    <form action="page.html" method="post">
      <label>
        Fruit:
        <input type="text" name="fruit" dirname="fruit-dir" value="cherry" />
      </label>
      <input type="submit" />
    </form>
  </body>
</html>
```

2.5.1 Input Validation

Some input types automatically validate format:

- `type="email"` → must contain @ and .
- `type="url"` → must start with `http://` or `https://`
- `type="number"` → must be numeric
- `type="date"` → must be a valid date

List of Validation Attributes:

- `required` → field must be filled

- min / max → numeric/date range
- minlength / maxlength → character length limits
- pattern → custom regex rule
- step → numeric intervals
- type → built-in format validation
- novalidate → disable validation

```

<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>HTML Form</title>
  </head>
  <body>
    <form action="page.html" method="post">
      <label>Email:</label>
      <input type="email" name="email" required><br/>
      <label>Age (18-60):</label>
      <input type="number" name="age" min="18" max="60" required><br/>
      <label>Username (5-10 chars):</label>
      <input type="text" name="username" minlength="5" maxlength="10" required><br/>
      <label>Only Letters:</label>
      <input type="text" name="letters" pattern="[A-Za-z]+" required><br/>
      <label>Even Number:</label>
      <input type="number" name="even" step="2" required><br/>
      <label>Website:</label><input type="url" name="website" required><br/>
      <input type="submit" />
    </form>
    <!-- <form novalidate>
      <input type="email" name="email" required>
      <input type="submit"></form> -->
  </body>
</html>

```

2.5.2 Accessibility and Usability

When including inputs, it is an accessibility requirement to add labels alongside. This is needed so those who use assistive technologies can tell what the input is for. Also, clicking or touching a label gives focus to the label's associated form control. This improves the accessibility and usability for sighted users, increases the area a user can click or touch to activate the form control. This is especially useful (and even needed) for radio buttons and checkboxes, which are tiny.

See above example under 2.4 and 2.5 how to associate the `<label>` with an `<input>` element.

2.6 What is Git and why use it?, Git setup and config (git config), Creating a local repository: git init

2.6.0 What is Git and why do we use it?

- Git is a **Version Control System (VCS)**, means Every developer has a full copy of repo.
- It helps keep track of every change you make to files in a project.
- If something breaks, you can go back to an earlier version.
- Multiple people can work on the same project without disturbing each other's work.
- Git is widely used in the software industry to manage projects and teamwork.

Key points:

- Git runs on your local computer.
- GitHub is a website (cloud) where Git projects can be stored and shared.
- Together, Git + GitHub = teamwork + backup + tracking changes.

Git has four important areas:

1. **Working Directory** – Where you actually edit files (your project folder).
2. **Staging Area (Index)** – A temporary area where you put the changes you want to commit.
3. **Local Repository** – The permanent history of commits stored on your computer.
4. **Remote Repository (GitHub/GitLab/Bitbucket)** – The copy of your repository stored on the server for sharing and collaboration.